



A Multi-attribute Quantum Decision Model Based on Grey Relational Analysis

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ABSTRACT

Aiming at the interference problem between decision-makers in uncertain group decision making, this paper proposes a new interference angle measure method based on quantum-like Bayesian network (QLBN). Firstly, the grey relational degree is extended to the positive and negative range, and then the grey relational degree is calculated according to the sequential growth trend, and it is transformed into the interference value. In this framework, the weights of the decision-makers are set as the first layer of QLBN, where they are determined based on the similarity between the personal opinion and the leading decision-maker's opinion, and the conditional probabilities are calculated by combining the attributes' weights and evaluation values. Finally, the quantum probabilities of alternatives are obtained. The alternatives are ranked according to the quantum probabilities, and the robustness and validity of the model are verified by numerical example. The result shows that the interference angle calculation method based on grey relational degree has a good performance in decision-making scenarios with incomplete information, and significantly improves the robustness and applicability of the quantum decision model.

Keywords: Multi-attribute group decision making, Grey relational degree, Quantum probability theory, Interference angle measure

1. Introduction

Multi-attribute group decision making (MAGDM) refers to the process of selecting or optimizing an alternative by comprehensively evaluating multiple interrelated attributes in the decision-making process involving multiple decision-makers^[1]. The MAGDM method is especially suitable for those situations where the decision objectives are multiple and there is a certain relationship between the attributes. Decision-makers often come from different fields or have different backgrounds, so their evaluation of each attribute may differ. MAGDM aims to make optimal or acceptable decisions in a complex multi-attribute, multi-objective environment by bringing together the opinions of decision-makers. The basic steps usually include the selection and definition of attributes, the evaluation of schemes, the weight allocation of decision-makers and the generation of final decisions^[2,3].

However, in practical applications, multi-attribute group decision making often faces the challenge of uncertain decision

environment^[4,5]. In many complex and dynamic decision problems, decision-makers cannot obtain complete and accurate decision information, or the decision environment itself is full of uncertainty and fuzziness. These uncertainties arise from a number of factors, such as the absence of data, vagueness of evaluation criteria, inconsistencies in expert assessments, and changes in the external environment. Under this background, the traditional deterministic decision-making method is difficult to deal with effectively. How to reasonably integrate the opinions of multiple parties and overcome the troubles caused by incomplete or fuzzy information has become an important problem in multi-attribute group decision-making. The uncertainty factor is particularly significant in the decision-making scene of disaster emergency rescue. Disaster events, such as earthquakes, floods, typhoons, etc., are often accompanied by suddenness, complexity, and unpredictability, which makes disaster emergency decision-making a highly challenging task. In these scenarios, decision-makers not only need to quickly access multiple sources of information from different domains, but also need to consider multiple interrelated attributes, such as disaster losses, rescue resources, evacuation routes, time constraints, and personnel safety.

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In terms of post-disaster resource allocation, emergency rescue team scheduling and post-disaster reconstruction, decision-makers must face such problems as incomplete information, fuzzy evaluation criteria and urgent time. Especially in high-risk and highly uncertain environments, timeliness and accuracy of decision-making are crucial, and MAGDM method is an effective tool to deal with such complex decision-making scenarios^[6]. For example, in disaster emergency rescue, decision-makers need to comprehensively consider the disaster situation in the disaster area, the needs of the affected people, the distribution of rescue teams and the strain of resources. In this multi-dimensional decision, how to make a quick and reasonable decision within a limited time needs to integrate the opinions of experts in different fields to achieve the best decision from multiple angles and levels. Therefore, MAGDM method can be widely used in the field of emergency response to help decision-makers make reasonable and effective judgments under high pressure environment^[7,8].

In an environment with incomplete information and high uncertainty, grey relational analysis method can effectively analyze variable relationships with a small amount of information, so it plays an important role in the research of multi-attribute decision making^[9]. The core idea of grey relational analysis is to measure the similarity by calculating the correlation between reference sequence and comparison sequence, so as to rank different factors or schemes. This method does not rely on a large number of sample data, but measures the strength of the variable relationship by constructing the similarity between the sequences, so that it has good applicability and accuracy even when the data acquisition is difficult. Since Deng^[10] introduced grey relational analysis, the method has gradually developed into a diverse and widely applied system analysis tool. Traditional research primarily revolves around absolute and relative relational degree models. The former emphasizes the numerical differences between sequences, while the latter focuses on the trend of changes in sequences, providing fundamental support for similarity measurement in both static and dynamic systems. However, traditional models are often limited by a local perspective, making it difficult to fully reveal the global structural similarities between sequences. Therefore, in recent years, many scholars have proposed a series of new grey relational models from a global perspective. In frequency domain feature extraction, Li et al.^[11] proposed a grey relational model based on discrete Fourier transform, effectively capturing periodic fluctuations and global dynamic patterns by fitting time series in the frequency domain. In geometric structure modeling, Zhang and Li^[12] constructed a grey angle cosine model based on generalized greyness of interval grey numbers, integrating proximity and similarity to enhance the structural correlation measurement of interval data. Xie and Liu^[13] introduced the concept of grey distance, embedding grey sequences into geometric space and achieving distance measurement of the overall form. In addressing the correlation of complex structured data, Liu et al.^[14] proposed a cross-sequence grey relational model based on differences, introducing the definitions of "cross-sequence" and "regional elements" to overcome distortion caused by positive and negative cancellation of integral terms. Hu and Xie^[15] further constructed a coupling grey relational model based on matrix norms, quantifying the structural features of binary indicator submatrices to achieve global measurement of structural coupling between attributes. Zhang et al.^[16] introduced gradients to measure the local similarity

between continuous functions. Through the application of double integration, local similarity was transformed into global similarity, which provides a new perspective for analyzing the similarity of behavior matrices. In terms of model stability and consistency, Zhang and Luo^[17] introduced Monte Carlo methods for unbiased estimation of generalized closeness, effectively addressing issues of sample replacement dependency and closeness variability in traditional models. Additionally, to meet the needs of small sample, multivariable, and ordered data regulation, Zhang et al.^[18] proposed a grey similarity model that combines ordered dynamic time warping with L1 norm and uses k-nearest neighbor algorithms to establish a risk warning mechanism for construction quality, demonstrating the new application of grey relational methods in complex sequence structure identification and classification.

In the research field of multi-attribute decision making, grey relational degree is gradually applied to identify and rank factor relationships in complex systems. In recent years, the application scope of grey relational has gradually expanded from the original traditional fields such as engineering, economy and management to such fields as emergency management, environmental assessment and medical decision-making, showing high reliability and practicability in many aspects^[19,20,21]. In addition, the calculation process of grey relational degree is simple and not limited by data scale, which makes its application in multi-attribute group decision making highly flexible^[22]. Grey relational degree reduces the interference of subjectivity on decision results and provides important support for scientific and objective analysis of decision variables. In group decision making, grey relational degree can effectively quantify the correlation between each decision-maker or attribute and the whole group result, which lays a solid foundation for further optimization of group decision making model. The traditional grey relational degree usually measures the strength of the relationship between variables by calculating the similarity between each series. The initial studies mostly focused on the positive relational case, that is, when the change trend of two variables is consistent, the relational degree is higher. However, with the complexity of decision requirements, the concept of negative association is gradually introduced to describe the inverse relationship between variables. When one variable increases while another variable decreases, negative relational can effectively depict this inverse relationship and expand the application scope of grey relational^[23,24]. Based on this, the value interval of grey relational degree is gradually extended to $[-1,0) \cup (0,1]$, where positive value represents positive relational, negative value represents reverse relational, and zero value indicates no significant relational between the two variables^[25]. This extension not only improves the flexibility of grey relational analysis, but also enhances its applicability in multi-attribute group decision making.

Quantum probability decision theory is an emerging framework of decision theory, which aims to explain and simulate complex decision behavior through quantum mechanics-like probability models^[26]. The "interference effect" introduced in this framework is one of the core concepts, which refers to the mutual interference among decision-makers, that is, the preference or judgment of decision-makers will be affected by the opinions of other decision-makers, resulting in the deviation of decision results^[27,28,29]. This effect is derived from the double slit experiment in quantum mechanics and is used in the field of decision making to

explain irrational behavior caused by incomplete or uncertain information^[30]. The interference effect reveals the interdependence between decision-makers' preferences, which is particularly common in group decision-making, especially when there is disagreement or information asymmetry, it is more likely to produce bias or inconsistency. Under the quantum-like Bayesian network (QLBN), there are various methods for calculating the interference angle, such as cosine similarity, Shannon entropy and Deng entropy^[31,32,33,34]. Cosine similarity is simple and easy to implement by calculating the cosine of the angle between probability vectors, but may lose accuracy in the case of uneven data distribution. Shannon entropy describes the interference effect through information uncertainty, which is suitable for the analysis of uniformly distributed events, but it cannot fully reflect the subtle influence of probability distribution difference on interference angle. On this basis, Deng entropy is further considered about the difference of probability distribution, but its complex calculation process increases the difficulty of model implementation, and the adaptability to a small number of samples is poor. Although these methods have their advantages, they all have some limitations, especially in complex and unevenly distributed situations, which may not fully reflect the complexity of interference effects.

The main motivations of this paper are as follows:

Motivation 1: Most traditional MAGDM models assume that decision-makers' judgments or preferences are independent of each other. However, in the actual disaster emergency and other complex situations, the opinions of decision-makers often affect each other, resulting in interference effects. This mutual interference may significantly affect the stability and consistency of decision results, which makes it difficult for traditional models to accurately depict the relationship between decision-makers.

Motivation 2: The measurement of the interference value of the decision-maker is a difficult problem, especially in determining the phase angle difference between two decision-makers. The current methods for calculating interference angle have some limitations, such as cosine similarity, Shannon entropy and Deng entropy, which may lose accuracy in the face of complex distribution or small number of samples. Cosine similarity is sensitive to data distribution, so it is difficult to deal with uneven preferences effectively. Shannon entropy is suitable for uniform distribution, but it is difficult to reflect preference differences in complex decision situations. Deng entropy is complex and sensitive to small samples, although distribution differences are taken into account.

Considering these motivations, a novel quantum group decision model is proposed to improve the accuracy and efficiency of decision making. Firstly, this paper constructs an optimization model based on grey relational degree to solve decision-maker weight. Secondly, a quantum-like Bayesian network is constructed to deal with interference effects among decision-makers. Among them, sgn function is combined to project the range of gray relational degree to the range $[-1,0) \cup (0,1]$ to measure the interference value among decision-makers. Finally, alternatives are sorted according to quantum probability.

The main contributions of this paper include:

1) In view of the limitations of the traditional multi-attribute group decision-making model, which assumes that decision-makers are independent of each other, this paper introduces the

concept of interference effect and proposes a model that can quantify the interaction between decision-makers. In this model, the interference effect is regarded as the mutual interference among decision-makers, which more truly portrays the interaction between group decision making, and improves the applicability and accuracy of the model in complex decision making situations.

2) Aiming at the shortcomings of existing interference angle calculation methods, this paper proposes a new interference angle calculation method based on grey relational degree. Different from the traditional cosine similarity, grey relational degree performs well in the case of uncertainty and incomplete information, and can adapt more flexibly to complex and unevenly distributed decision scenarios. This method not only improves the accuracy of interference angle calculation, but also reduces the dependence on data distribution uniformity, and provides a more adaptive and robust approach to mutual interference measurement in multi-attribute group decision making.

3) The weights of decision-makers are determined according to the similarity between the evaluation of each decision-maker and the evaluation of the leading decision-maker, which provides support for the probability of key nodes in quantum-like Bayesian networks.

This paper is organized as follows: Section 2 reviews the framework of quantum probability theory. In Section 3, a multi-attribute quantum group decision model based on grey relational analysis is proposed. Section 4 verifies the stability and robustness of the model based on a disaster emergency rescue example, and verifies the effectiveness of the model through comparative analysis. Finally, Section 5 summarizes the main findings and contributions of this study, and puts forward prospects for future research directions.

2. Introduction

2.1. Quantum probability theory

Unlike classical probability theory, in quantum probability theory, all possible events can be treated as subspaces of Hilbert Spaces.

Definition 1^[28]. Hilbert space is a generalization of Euclidean space, which is no longer limited to the finite dimensional case. Let H be a real linear space, if any two vectors x^0 and y^0 in H correspond to a real number, denoted as (x^0, y^0) , the following conditions are met:

1) For any two vectors x^0 and y^0 in H , there is $(x^0, y^0) = (y^0, x^0)$.

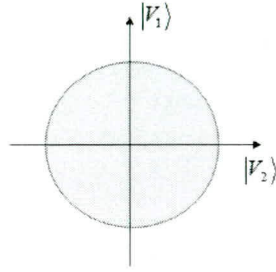
2) For any three vectors x^0, y^0, z^0 and the crisp numbers α^0, β in H , there is $(\alpha^0 x^0 + \beta y^0, z^0) = \alpha^0 (x^0, y^0) + \beta (y^0, z^0)$.

3) For every vector x^0 in H , there is $(x^0, x^0) \geq 0$, and the necessary and sufficient condition for $(x^0, x^0) = 0$ is $x^0 = 0$. Then (x^0, y^0) is defined to be an inner product on H , and H is defined the inner product space.

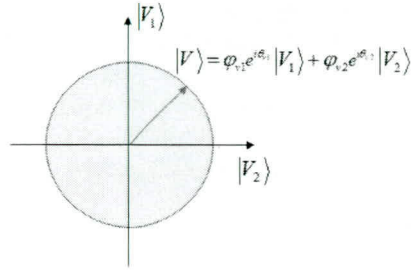
Constitutes a complex vector space spanned by a set of orthogonal basis vectors $V = \{|V_1\rangle, \dots, |V_n\rangle\}$. In Hilbert space, all events are defined in terms of quantum superposition states $|D\rangle$, which can be represented by linear combinations of orthogonal basis vectors, as shown in Figure 1.

$$|D\rangle = \langle V_1 | D \rangle |V_1\rangle + \langle V_2 | D \rangle |V_2\rangle + \dots + \langle V_n | D \rangle |V_n\rangle$$

$$\begin{aligned}
&= \psi_1 |V_1\rangle + \psi_2 |V_2\rangle + \cdots + \psi_n |V_n\rangle \\
&= \varphi_1 e^{i\theta_1} |V_1\rangle + \varphi_2 e^{i\theta_2} |V_2\rangle + \cdots + \varphi_n e^{i\theta_n} |V_n\rangle.
\end{aligned} \quad (1)$$



(a) The Hilbert space in QPT



(b) An event in the Hilbert space

Figure 1 Events in Hilbert space

3. Multi-attribute quantum decision model based on grey relational analysis

The main goal of a MAGDM problem is to derive a consensus-based solution from multiple alternatives with multiple attributes. The typical elements of MAGDM are as follows.

1) There are $s+1$ decision-makers $D = \{d_0, d_1, d_2, \dots, d_s\}$ ($s \geq 3$), where d_0 stands for leading decision-maker and d_k ($k=1, 2, \dots, s$) stands for following decision-maker. The weights set of decision-makers is $\Lambda = \{\lambda_0, \lambda_1, \lambda_2, \dots, \lambda_s\}$, and the conditions $\lambda_k \geq 0$ ($k=0, 1, 2, \dots, s$) and satisfy that the sum of weights is 1.

2) The limited set of alternative alternatives is $A = \{a_1, a_2, \dots, a_m\}$ ($m \geq 2$), which are potential solutions to the multi-attribute group decision problem.

3) The attribute set is $C = \{c_1, c_2, \dots, c_n\}$ ($n \geq 2$), the attributes' weights set is $W^k = \{\omega_1^k, \omega_2^k, \dots, \omega_n^k\}$, conditions $\omega_j^k \geq 0$ ($j=1, 2, \dots, n, k=1, 2, \dots, s$) and satisfy that the sum of weights is 1.

3.1. A novel positive and negative grey relational degree

Definition 2. Suppose there are two sequences X_k ($k=1, 2, \dots, s$) and X_h ($h=1, 2, \dots, s$) as follows:

$$X_k = \begin{bmatrix} x_k(1) \\ x_k(2) \\ \vdots \\ x_k(m) \end{bmatrix}, X_h = \begin{bmatrix} x_h(1) \\ x_h(2) \\ \vdots \\ x_h(m) \end{bmatrix}, k, h = 1, 2, \dots, s.$$

And the corresponding difference vectors are ΔX and ΔY as follows:

$$\Delta X_k = \begin{bmatrix} x_k(2) - x_k(1) \\ x_k(3) - x_k(2) \\ \vdots \\ x_k(m) - x_k(m-1) \end{bmatrix}, \Delta X_h = \begin{bmatrix} x_h(2) - x_h(1) \\ x_h(3) - x_h(2) \\ \vdots \\ x_h(m) - x_h(m-1) \end{bmatrix}, k, h = 1, 2, \dots, s.$$

Then the grey relational coefficients of the two vectors are defined as:

$$\begin{aligned}
\varepsilon(\Delta x_k(i), \Delta x_h(i)) &= \text{sgn}(\Delta) \cdot \frac{\min |\Delta x_k(i) - \Delta x_h(i)| + \alpha \max |\Delta x_k(i) - \Delta x_h(i)|}{|\Delta x_k(i) - \Delta x_h(i)| + \alpha \max |\Delta x_k(i) - \Delta x_h(i)|}, \\
i &= 1, 2, \dots, m, k = 1, 2, \dots, s, h = 1, 2, \dots, s.
\end{aligned} \quad (2)$$

Where, $\alpha \in [0, 1]$ represents the distinguish coefficient. sgn function is introduced to reflect the positivity and negativity between two vectors, thus conforming to the interval range of interference angle cosine value, sgn function is defined as follows:

$$\text{sgn}(\Delta) = \begin{cases} 1, & \Delta x_k(i) \cdot \Delta x_h(i) \geq 0, \\ -1, & \Delta x_k(i) \cdot \Delta x_h(i) < 0 \end{cases} \quad (3)$$

Theorem 1. The range of the grey relational coefficient $\varepsilon(\Delta x_k(i), \Delta x_h(i))$ is $[-1, 0) \cup (0, 1]$.

Proof. Firstly, the value range of the function $\text{sgn}(\Delta)$ is $\{-1, 1\}$. Secondly, since $\alpha \in [0, 1]$, $\frac{\min |\Delta x_k(i) - \Delta x_h(i)| + \alpha \max |\Delta x_k(i) - \Delta x_h(i)|}{|\Delta x_k(i) - \Delta x_h(i)| + \alpha \max |\Delta x_k(i) - \Delta x_h(i)|}$ belongs to the interval $(0, 1]$, when $\Delta x_k(i) \cdot \Delta x_h(i) \geq 0$, the grey relational coefficient $\varepsilon(\Delta x_k(i), \Delta x_h(i))$ is positive; when $\Delta x_k(i) \cdot \Delta x_h(i) < 0$, the grey relational coefficient $\varepsilon(\Delta x_k(i), \Delta x_h(i))$ is negative. So combining the range of the numerator and denominator of Eq. (2), we can get the range of the grey relational coefficient $\varepsilon(\Delta x_k(i), \Delta x_h(i))$ is $[-1, 0) \cup (0, 1]$. $\square$

Theorem 2. The grey relational coefficient $\varepsilon(\Delta x_k(i), \Delta x_h(i))$ satisfies even symmetry.

Proof. Consider the calculation formula for the grey relational coefficient:

$$\varepsilon(\Delta x_k(i), \Delta x_h(i)) = \text{sgn}(\Delta) \cdot \frac{\min |\Delta x_k(i) - \Delta x_h(i)| + \alpha \max |\Delta x_k(i) - \Delta x_h(i)|}{|\Delta x_k(i) - \Delta x_h(i)| + \alpha \max |\Delta x_k(i) - \Delta x_h(i)|}, \text{ where}$$

the $\text{sgn}(\Delta)$ depends on the product of $\Delta x_k(i)$ and $\Delta x_h(i)$. When exchanging $\Delta x_k(i)$ and $\Delta x_h(i)$, the sign function remains unchanged because of the commutative property of multiplication. Additionally, the symmetry of the absolute value function ensures that both the minimum and maximum terms remain invariant. Therefore, the entire expression remains unchanged when exchanging $\Delta x_k(i)$ and $\Delta x_h(i)$,

$\varepsilon(\Delta x_k(i), \Delta x_h(i)) = \varepsilon(\Delta x_h(i), \Delta x_k(i))$. Thus, the grey relational coefficient satisfies even symmetry. $\square$

Theorem 3. The grey relational coefficient $\varepsilon(\Delta x_k(i), \Delta x_h(i))$ satisfies proximity.

Proof. If $\Delta x_k(i)$ and $\Delta x_h(i)$ are close, $|\Delta x_k(i) - \Delta x_h(i)| \rightarrow 0$, then the minimum term in the numerator of the grey relational coefficient: $\min |\Delta x_k(i) - \Delta x_h(i)| \rightarrow 0$. The term $|\Delta x_k(i) - \Delta x_h(i)|$ in the denominator also tends to zero. Therefore, the entire fraction approaches 1. Additionally, since the $\text{sgn}(\Delta)$ depends on the signs of $\Delta x_k(i)$ and $\Delta x_h(i)$, and both values have the same sign, the sign function remains unchanged. As a result, the grey relational coefficient $\varepsilon(\Delta x_k(i), \Delta x_h(i))$ approaches 1 when $\Delta x_k(i)$ and $\Delta x_h(i)$ are close. Thus, the grey relational coefficient satisfies proximity. $\square$

On the basis, the grey relational degree is calculated as follows:

$$\tilde{\varepsilon}_{x_k, x_h} = \frac{1}{m} \sum_{i=1}^m \varepsilon(\Delta x_k(i), \Delta x_h(i)), k=1, 2, \dots, s, h=1, 2, \dots, s. \quad (4)$$

3.2. Decision-making information collection and processing

For each decision-maker $d_k (k=0, 1, 2, \dots, s)$, the evaluation information of alternative $a_i (i=1, 2, \dots, m)$ about attribute $c_j (j=1, 2, \dots, n)$ is represented by the crisp number $z_{ij}^k (i=1, 2, \dots, m, j=1, 2, \dots, n, k=1, 2, \dots, s)$, and the evaluation information matrix is represented by $Z^k = (z_{ij}^k)_{m \times n} (k=0, 1, 2, \dots, s)$, that is:

$$Z^k = \begin{pmatrix} z_{11}^k & z_{12}^k & \dots & z_{1n}^k \\ z_{21}^k & z_{22}^k & \dots & z_{2n}^k \\ \vdots & \vdots & \ddots & \vdots \\ z_{m1}^k & z_{m2}^k & \dots & z_{mn}^k \end{pmatrix}, k=0, 1, 2, \dots, s. \quad (5)$$

In MAGDM problems, because different attributes usually have different dimensions and value ranges, direct use of raw data for analysis may lead to weight imbalance between attributes, and then affect the fairness and science of decision results. Therefore, it is necessary to normalize the evaluation information matrix, and to eliminate the dimensional differences between attributes by converting data with different attributes into a unified scale.

$$\tilde{z}_{ij}^k = \begin{cases} \frac{z_{ij}^k - \min_i z_{ij}^k}{\max_i z_{ij}^k - \min_i z_{ij}^k}, & c_j \text{ for benefit,} \\ \frac{\max_i z_{ij}^k - z_{ij}^k}{\max_i z_{ij}^k - \min_i z_{ij}^k}, & c_j \text{ for cost,} \end{cases} \quad (6)$$

$i=1, 2, \dots, m, j=1, 2, \dots, n, k=0, 1, 2, \dots, s.$

3.3. Determination of the weights of decision-makers based on group consensus

In MAGDM problems, the consistency of group opinions is of great significance to improve the rationality and effectiveness of decision making. Because of the difference of knowledge background, preference and experience of different decision-makers, it often leads to the inconsistency of individual opinions. In order to assign the weight of decision-makers more scientifically in the decision-making process, the weight of each decision-maker can be calculated according to its proximity to the opinion of the leading decision-maker.

Firstly, the comprehensive index values of each decision-maker for each alternative are:

$$Q^k(a_i) = \sum_{j=1}^n \varpi_j^k \tilde{z}_{ij}^k, i=1, 2, \dots, m, k=1, 2, \dots, s. \quad (7)$$

Secondly, the comprehensive value of leading decision-maker for each alternative is:

$$Q^0(a_i) = \sum_{j=1}^n \varpi_j^0 \tilde{z}_{ij}^0, i=1, 2, \dots, m. \quad (8)$$

Where, ϖ_j^0 represents the attributes' weights of lead decision-maker d_0 .

On this basis, the grey relational coefficient of each decision-maker's opinion and leading decision-maker's opinion in each alternative is as follows:

$$\xi(Q^k(a_i), Q^0(a_i)) = \frac{\min_i \min_k |Q^k(a_i) - Q^0(a_i)| + \alpha \max_k \max_i |Q^k(a_i) - Q^0(a_i)|}{|Q^k(a_i) - Q^0(a_i)| + \alpha \max_k \max_i |Q^k(a_i) - Q^0(a_i)|}, \quad (9)$$

where, $\alpha (0 < \alpha < 1)$ is the distinguishing coefficient.

The relational between each decision-maker and leading decision-maker's opinion is defined as:

$$\gamma_{k0} = \frac{1}{m} \sum_{i=1}^m \xi(Q^k(a_i), Q^0(a_i)), k=1, 2, \dots, s. \quad (10)$$

Definition 3. Let $\eta_k = 1 - \gamma_{k0}$ be the correction degree of the follower decision-maker $d_k (k=1, 2, \dots, s)$ under the influence of the leading decision-maker d_0 , and $\eta_k (k=1, 2, \dots, s)$ belongs to $(0, 1]$.

Corollary 1. When $\gamma_{k0} = 1$, that is, the opinions of the following decision-maker $d_k (k=1, 2, \dots, s)$ and the leading decision-maker d_0 converge, and the correction degree $\eta_k = 1$. When γ_{k0} is approximately equal to 0, that is, the opinions of $d_k (k=1, 2, \dots, s)$, the following decision-maker, and d_0 , the leading decision-maker, are basically inconsistent, and the correction degree $\eta_k (k=1, 2, \dots, s)$ is approximately 1.

In MAGDM, correction degree $\eta_k (k=1, 2, \dots, s)$ can be further used to calculate decision-makers' weights. Since the influence of decision-makers is often related to the stability of their opinions and the degree of group consensus, decision-makers with lower corrections (that is, those who are close to the leading decision-maker) tend to have higher weights, while decision-makers with higher corrections (that is, those who have a large deviation from the leading decision-maker) may be given lower weights. On this basis, the weights of each decision-maker are normalized, and the weight of leading decision-maker and following decision-makers are obtained as follows:

$$\lambda_k = \begin{cases} \frac{1}{1 + \sum_{k=1}^s \left(\frac{1}{1 + \eta_k} \right)}, & k=0, \\ \frac{1/(1 + \eta_k)}{1 + \sum_{k=1}^s \left(\frac{1}{1 + \eta_k} \right)}, & k=1, 2, \dots, s. \end{cases} \quad (11)$$

3.4. Calculation of quantum probabilities for each alternative

3.4.1. Construction of quantum-like Bayesian network

In the process of multi-attribute group decision making, this paper considers the opinion interference among decision-makers, and proposes a model to solve the interference effect based on QLBN. The first layer of the framework is the decision-maker layer, and the second layer is the alternative layer. The model further

improves the decision-making process by introducing the opinion interference mechanism among decision-makers, as follows:

Firstly, the weights of decision-makers are transformed into the first layer of quantum-like Bayesian network.

$$P(d_k) = \lambda_k, k = 0, 1, 2, \dots, s. \quad (12)$$

Secondly, the evaluation matrix and attributes' weights of each decision-maker are aggregated and converted into conditional probabilities.

$$P(a_i | d_k) = \frac{\sum_{j=1}^n \omega_j^k \bar{z}_{ij}^k}{\sum_{i=1}^m \sum_{j=1}^n \omega_j^k \bar{z}_{ij}^k}, i = 1, 2, \dots, m, k = 0, 1, 2, \dots, s. \quad (13)$$

3.4.2. Interference angle measure based on grey relational degree

In the group decision making process, the leading decision-maker usually plays a central role, and his opinions and behaviours have a significant influence on the decision result. Therefore, in the framework of this section, the main focus is on the interference effect between the leading decision-maker and the following decision-maker. This interference effect is reflected in the direct or indirect influence of the opinions, preferences or decision-making tendencies of the leading decision-makers on the judgment and behaviour of the following decision-makers. In contrast, the mutual interference effect between the following decision-makers has a relatively weak influence on the overall decision result, so it is ignored. This hypothesis not only simplifies the complexity of the model, but also highlights the dominant position of the leading decision-maker in the group decision making, thus revealing the dynamic relationship between the leading decision-maker and the following decision-maker and the mechanism of action on the decision result.

Firstly, construct the vectors $\bar{v}_k (k = 1, 2, \dots, s)$ and $\bar{q}_0$:

$$\bar{v}_k = \begin{bmatrix} P(d_k)p(a_1 | d_k) \\ P(d_k)p(a_2 | d_k) \\ \vdots \\ P(d_k)p(a_m | d_k) \end{bmatrix}, \bar{q}_0 = \begin{bmatrix} P(d_0)p(a_1 | d_0) \\ P(d_0)p(a_2 | d_0) \\ \vdots \\ P(d_0)p(a_m | d_0) \end{bmatrix}, k = 1, 2, \dots, s. \quad (14)$$

Secondly, the difference sequence vectors $\bar{\gamma}_k (k = 1, 2, \dots, s)$ and $\bar{\tau}_0$ are constructed:

$$\bar{\gamma}_k = \begin{bmatrix} P(d_k)p(a_2 | d_k) - P(d_k)p(a_1 | d_k) \\ P(d_k)p(a_3 | d_k) - P(d_k)p(a_2 | d_k) \\ \vdots \\ P(d_k)p(a_m | d_k) - P(d_k)p(a_{m-1} | d_k) \end{bmatrix}, \quad (15)$$

$$\bar{\tau}_0 = \begin{bmatrix} P(d_0)p(a_2 | d_0) - P(d_0)p(a_1 | d_0) \\ P(d_0)p(a_3 | d_0) - P(d_0)p(a_2 | d_0) \\ \vdots \\ P(d_0)p(a_m | d_0) - P(d_0)p(a_{m-1} | d_0) \end{bmatrix}, k = 1, 2, \dots, s.$$

In this paper, according to the growth or decline of the trend of sequence change, we judge whether the series are positively correlated or negatively correlated. According to this, the difference between the points is used to define the trend sequence, the correlation sign is defined according to the consistency of the difference sign, and the relational degree is defined according to the proximity of the trend sequence.

On this basis, the grey relational degree is used to measure the interference angle, which is mainly based on the core idea of grey system theory, that is, in the case of incomplete information or uncertain data, the mutual influence relationship is described by the similarity between known variables. Grey relational analysis is especially suitable for small samples, incomplete information and high system complexity, which is in line with the background of quantum group decision making. Compared with cosine similarity [35], grey relational degree not only focuses on the difference of evaluation angle of decision-makers in a certain scheme, but also further describes the relative change trend of evaluation value itself. In addition, compared with the entropy method, the calculation of grey relational degree does not depend on the probability distribution and does not need to set the calculation method of information entropy in advance, so it can still maintain good stability and applicability when the data scale is small. Entropy method [33] can effectively measure uncertainty in the case of sufficient information, but when there is less data or the evaluation of decision-makers is close, its calculation results may be distorted, while grey relational analysis can still maintain a strong discrimination ability under limited data. Therefore, the method proposed in this paper to measure interference values based on grey relational degree not only provides a more data-driven way to measure interference effects among decision-makers, but also enhances the robustness of the model, making the quantum group decision method more adaptable and computationally rational in complex and uncertain environments.

According to Definition 2, the grey relational coefficient is calculated. Therefore, the grey relational degree is calculated as follows:

$$\tilde{e}_{k0} = \frac{1}{m-1} \sum_{i=2}^m \varepsilon(\gamma_{ki}, \tau_{0i}), k = 1, 2, \dots, s. \quad (16)$$

Although the value range of the novel grey relational degree proposed in this paper does not include zero and cannot represent the complete independence of decision-makers, this setting is reasonable and practical under the background of group decision making with high uncertainty. The assumption of complete independence among decision-makers means that in all cases the preferences, judgments, and choices of decision-makers do not influence each other. In practical decision-making, however, complete independence is almost impossible. Decision-makers' preferences are influenced not only by personal experience, knowledge and cognitive biases, but also by the external environment and the opinions of other decision-makers. In the uncertain decision-making environment with incomplete information, environmental changes or individual cognitive differences, the mutual influence and interference between decision-makers is the normal. The measure of interference intensity based on grey relational is precisely to capture this weak and complex interaction. Even if the opinions of decision-makers are close or converge, there is still a certain degree of interference effect. Therefore, the cosine value of the interference angle is not zero, helping to avoid oversimplified assumptions and ensuring that the model can truly reflect the weak interference and interaction between decision-makers. In group decision-making, decision-makers usually influence each other, and even when some decision-makers have similar preferences, their decisions will still be subtly influenced by each other's opinions. Therefore, the use of

grey relational degree to measure the interference intensity between decision-makers is still feasible and appropriate, although it cannot represent a completely independent situation.

Therefore, the cosine of the interference angle is obtained.

$$\cos(\theta_{k0}) = \tilde{\varepsilon}_{k0}, k = 1, 2, \dots, s. \quad (17)$$

Finally, the quantum probability values of each alternative are calculated.

$$P(a_i) = \frac{1}{\sigma} \left[\frac{\sum_{k=1}^s P(d_0)P(a_i|d_0)P(d_k)P(a_i|d_k) + 2 \sum_{k=1}^s \sqrt{P(d_k)P(a_i|d_k)P(d_0)P(a_i|d_0)} \cos(\theta_{k0})}{\sum_{k=1}^s P(d_0)P(a_i|d_0)P(d_k)P(a_i|d_k) + 2 \sum_{k=1}^s \sqrt{P(d_k)P(a_i|d_k)P(d_0)P(a_i|d_0)} \tilde{\varepsilon}_{k0}} \right], \quad (18)$$

$$i = 1, 2, \dots, m.$$

where, the normalization factor σ is defined as:

$$\sigma = \frac{1}{m} \left[\frac{\sum_{k=1}^s P(d_0)P(a_i|d_0)P(d_k)P(a_i|d_k) + 2 \sum_{k=1}^s \sqrt{P(d_k)P(a_i|d_k)P(d_0)P(a_i|d_0)} \tilde{\varepsilon}_{k0}}{\sum_{k=1}^s P(d_0)P(a_i|d_0)P(d_k)P(a_i|d_k) + 2 \sum_{k=1}^s \sqrt{P(d_k)P(a_i|d_k)P(d_0)P(a_i|d_0)} \tilde{\varepsilon}_{k0}} \right]. \quad (19)$$

3.5. Decision steps

Step 1: Collect the evaluation information matrix $Z^k (k = 1, 2, \dots, s)$ of each decision-maker, and obtain the normalized matrix $\tilde{Z}^k (k = 1, 2, \dots, s)$ of each decision-maker according to Eq. (6).

Step 2: Determine the weights of decision-makers based on grey relational degree. According to Eqs. (7-10), the grey relational degrees are calculated, and on this basis, the decision-makers' weights are obtained by Eq. (11).

Step 3: Construct a quantum-like Bayesian network. The decision-makers' weights are the first layer of the QLBN, and the conditional probabilities are calculated according to Eq. (13) as the second layer network.

Step 4: Measure the interference value between decision-makers according to Eqs. (14-17), calculate the quantum probability of each alternative through Eqs. (18-19) and rank it.

4. Case analysis

In Huludao, Liaoning Province, China, sustained heavy rainfall caused severe flooding in 2024, causing damage to roads, flooding of farmland and flooding of residential areas, which had a serious impact on local traffic, agriculture and residents' lives. Faced with the disaster, the local government quickly convened a decision-making group of three decision-makers to assess emergency alternatives. Decision-maker d_0 is a senior expert in the field of emergency rescue. Decision-maker d_1 is an expert in disaster management, focusing on resettlement and infrastructure rehabilitation. Decision-maker d_2 is a traffic engineering expert, focusing on the speed of traffic recovery. Decision-maker d_3 is an agricultural technologist who focuses on the effectiveness of

agricultural recovery. The three decision-makers will evaluate the options in their respective fields to determine the most optimal disaster relief and recovery alternative. In the MAGDM, five attributes will be used as the evaluation attribute for the alternative, the weights set for the attributes is $W^k = \{0.2, 0.2, 0.2, 0.2, 0.2\}$, $k = 0, 1, 2, 3$. And the score range of each attribute is 0 to 10. The higher the score value, the greater the superiority of the scheme on the attribute.

1) Personnel security degree c_1 . This attribute measures the effectiveness of the program in protecting the lives of residents. The higher the score, the more effective the scheme is in terms of evacuation, resettlement and prevention of secondary disasters. The score ranges from 0 to 10. 0 indicates that the safety of personnel is not ensured and serious security risks exist. 10 indicates that comprehensive and rapid personnel resettlement can effectively ensure the safety of residents.

2) Traffic recovery speed c_2 . This attribute measures how quickly the program repairs damaged transportation infrastructure and how well traffic are restored. The higher the score, the faster the solution can restore access to major roads to support relief teams and the delivery of supplies. The score ranges from 0 to 10. 0 indicate that the traffic disruption cannot be restored. 10 indicates that all major roads have been repaired quickly and traffic is fully restored.

3) Improved flood control capacity c_3 . This attribute measures the effectiveness of the scheme in enhancing the resilience of the area to flood protection, focusing on the effect of strengthening flood defences and improving drainage systems. The higher the score, the more significant the improvement effect of flood control capacity. The score ranges from 0 to 10, where 0 indicates no flood control improvement and there is a risk of flooding again. 10 indicates that flood control facilities have been comprehensively strengthened and have strong disaster resistance.

4) Agricultural recovery effect c_4 . This attribute measures the program's effectiveness in helping to drain farmland and reduce agricultural losses. The higher the score, the more quickly the program can remove water from farmland, reduce crop losses, and help agriculture return to production. The score ranges from 0 to 10, where 0 indicates failure to provide agricultural assistance and continued flooding of farmland; 10 means rapid drainage and effective agricultural assistance to ensure the recovery of farmland and crops.

5) Efficiency of infrastructure restoration c_5 . This attribute measures the efficiency of the scheme in restoring infrastructure such as water supply and electricity. The higher the score, the faster the plan can restore the facilities needed for daily life and reduce the impact of disasters. The score ranges from 0 to 10. 0 indicates that infrastructure such as water supply and electricity are completely disrupted. 10 indicates that facilities have been repaired quickly and residents' life has completely returned to normal.

4.1. Decision steps

Step 1: The evaluation information matrix of each decision-maker is collected as follows:

$$Z^0 = \begin{bmatrix} 6 & 7 & 5 & 3 & 6 \\ 5 & 4 & 6 & 6 & 7 \\ 3 & 6 & 7 & 7 & 5 \\ 4 & 4 & 8 & 4 & 3 \\ 2 & 3 & 4 & 5 & 8 \end{bmatrix}, Z^1 = \begin{bmatrix} 3 & 4 & 3 & 4 & 7 \\ 6 & 8 & 4 & 6 & 6 \\ 5 & 1 & 7 & 8 & 5 \\ 4 & 5 & 5 & 2 & 3 \\ 1 & 9 & 3 & 1 & 4 \end{bmatrix},$$

$$Z^2 = \begin{bmatrix} 9 & 5 & 8 & 6 & 8 \\ 4 & 4 & 5 & 4 & 4 \\ 6 & 3 & 2 & 7 & 6 \\ 3 & 7 & 3 & 3 & 1 \\ 7 & 1 & 4 & 5 & 5 \end{bmatrix}, Z^3 = \begin{bmatrix} 7 & 8 & 6 & 2 & 4 \\ 2 & 6 & 5 & 8 & 8 \\ 6 & 4 & 7 & 3 & 1 \\ 9 & 2 & 2 & 4 & 9 \\ 4 & 7 & 4 & 6 & 5 \end{bmatrix}.$$

On this basis, normalization is carried out according to Eq. (6).

$$\tilde{Z}^0 = \begin{bmatrix} 1.0000 & 1.0000 & 0.2500 & 1.0000 & 0.6000 \\ 0.7500 & 0.2500 & 0.5000 & 0.2500 & 0.8000 \\ 0.2500 & 0.7500 & 0.7500 & 0.0000 & 0.4000 \\ 0.5000 & 0.2500 & 1.0000 & 0.7500 & 0.0000 \\ 0.0000 & 0.0000 & 0.0000 & 0.5000 & 1.0000 \end{bmatrix},$$

$$\tilde{Z}^1 = \begin{bmatrix} 0.4000 & 0.3750 & 0.0000 & 0.5714 & 1.0000 \\ 1.0000 & 0.8750 & 0.2500 & 0.2857 & 0.7500 \\ 0.8000 & 0.0000 & 1.0000 & 0.0000 & 0.5000 \\ 0.6000 & 0.5000 & 0.5000 & 0.8571 & 0.0000 \\ 0.0000 & 1.0000 & 0.0000 & 1.0000 & 0.2500 \end{bmatrix},$$

$$\tilde{Z}^2 = \begin{bmatrix} 1.0000 & 0.6667 & 1.0000 & 0.2500 & 1.0000 \\ 0.1667 & 0.5000 & 0.5000 & 0.7500 & 0.4286 \\ 0.5000 & 0.3333 & 0.0000 & 0.0000 & 0.7143 \\ 0.0000 & 1.0000 & 0.1667 & 1.0000 & 0.0000 \\ 0.6667 & 0.0000 & 0.3333 & 0.5000 & 0.5714 \end{bmatrix},$$

$$\tilde{Z}^3 = \begin{bmatrix} 0.7143 & 1.0000 & 0.8000 & 1.0000 & 0.3750 \\ 0.0000 & 0.6667 & 0.6000 & 0.0000 & 0.8750 \\ 0.5714 & 0.3333 & 1.0000 & 0.8333 & 0.0000 \\ 1.0000 & 0.0000 & 0.0000 & 0.6667 & 1.0000 \\ 0.2857 & 0.8333 & 0.4000 & 0.3333 & 0.5000 \end{bmatrix}.$$

Step 2: Determine the weights of decision-makers based on grey relevance idea. The weights of decision-makers are calculated according to Eqs. (7-11).

$$\Lambda = \{0.5069, 0.1142, 0.1561, 0.2228\}.$$

Step 3: Construct a quantum-like Bayesian network. The first layer is the weights of the decision-makers, and the conditional probability is calculated according to Eq. (13).

$$P(a_1 | d_0) = 0.3068, P(a_1 | d_1) = 0.1875, P(a_1 | d_2) = 0.3251, P(a_1 | d_3) = 0.2821,$$

$$P(a_2 | d_0) = 0.2032, P(a_2 | d_1) = 0.2526, P(a_2 | d_2) = 0.1947, P(a_2 | d_3) = 0.1553,$$

$$P(a_3 | d_0) = 0.1713, P(a_3 | d_1) = 0.1838, P(a_3 | d_2) = 0.1285, P(a_3 | d_3) = 0.1986,$$

$$P(a_4 | d_0) = 0.1992, P(a_4 | d_1) = 0.1963, P(a_4 | d_2) = 0.1798, P(a_4 | d_3) = 0.1934,$$

$$P(a_5 | d_0) = 0.1195, P(a_5 | d_1) = 0.1798, P(a_5 | d_2) = 0.1719, P(a_5 | d_3) = 0.1706.$$

Step 4: Interference angle measurement based on grey relational analysis. Where, the interference value between the following decision-maker and the leading decision-maker is calculated according to Eq. (14-17).

$$\tilde{\epsilon}_{10} = 0.5075, \tilde{\epsilon}_{13} = 0.7280, \tilde{\epsilon}_{23} = -0.0876.$$

Step 5: Calculate the quantum probability according to Eqs. (18-19):

$$P(a_1) = 0.2979, P(a_2) = 0.2114, P(a_3) = 0.1565,$$

$$P(a_4) = 0.1923, P(a_5) = 0.1419.$$

On this basis, the ranking is made according to the quantum probability, and the ranking result is as follows:

$$a_1 \succ a_2 \succ a_4 \succ a_3 \succ a_5.$$

4.2. Comparative analysis

In order to evaluate the effectiveness and robustness of the proposed model, numerical experiments will be conducted to compare and analyse with other existing decision models. The final ranking results obtained by different methods are shown in Figure. 2 and Table 1.

It can be seen from the analysis results in Table 2 that the scheme ranking results obtained by the proposed method are basically consistent with those of several existing decision models, which further verifies the reliability of the method. Figure 2 visually shows the correlation between the proposed method and other methods through Spearman rank correlation analysis. The results show that the Spearman correlation coefficient between the proposed method and other methods is generally high, indicating that there is a significant dependence and consistency between the ranking results.

Although traditional decision-making models such as TOPSIS [36] and TODIM [37] show strong applicability in multi-attribute decision-making problems, their core assumptions are

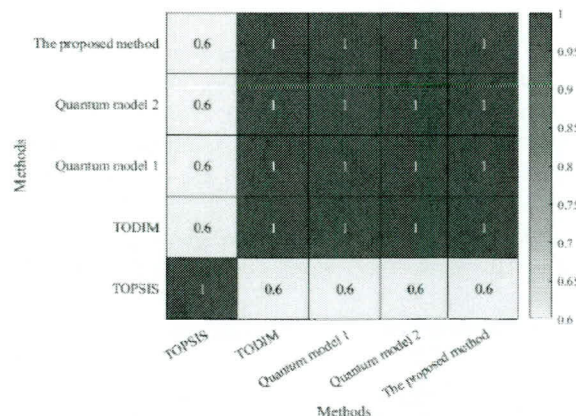


Figure 2 Spearman rank correlation of different methods

based on the independence of decision-makers and fail to fully consider the possible interference effects between decision-makers. This assumption may be true in simple decision-making scenarios, but in the complex actual decision-making environment, the mutual influence between decision-makers often cannot be ignored. Quantum model 1 [38] attempts to introduce quantum probability theory into the multi-attribute group decision process and provides a new perspective for decision analysis. However, it only analyses

from a specific interference perspective and fails to quantify specific interference values, which limits its operability in practical applications. Quantum model 2 [35] measures interference angle through cosine similarity, which makes up for the shortcomings of quantum model 1 to a certain extent, but cosine similarity method is more suitable for scenes with uniform data distribution, and its effectiveness may be limited in the face of high uncertainty and complexity of decision-making environment.

Table 1 Comparison of numerical experimental ranking results of different models.

Method	Ranking indices	Ranking
TOPSIS [36]	$C_1^+ = 0.8200, C_2^+ = 0.5003,$ $C_3^+ = 0.4456, C_4^+ = 0.5012,$ $C_5^+ = 0.3534.$	$a_1 > a_2 > a_3 > a_4 > a_5$
TODIM [37]	$\phi(a_1) = 1.8574, \phi(a_2) = -0.8816,$ $\phi(a_3) = -2.5018, \phi(a_4) = -1.7953,$ $\phi(a_5) = -3.0345.$	$a_1 > a_2 > a_4 > a_3 > a_5$
Quantum model 1 [38]	$P(a_1) = 0.2912, P(a_2) = 0.1991,$ $P(a_3) = 0.1708, P(a_4) = 0.1952,$ $P(a_5) = 0.1436.$	$a_1 > a_2 > a_4 > a_3 > a_5$
Quantum model 2 [35]	$P(a_1) = 0.2584, P(a_2) = 0.1989,$ $P(a_3) = 0.1825, P(a_4) = 0.1936,$ $P(a_5) = 0.1666.$	$a_1 > a_2 > a_4 > a_3 > a_5$
The proposed model	$P(a_1) = 0.2979, P(a_2) = 0.2114,$ $P(a_3) = 0.1565, P(a_4) = 0.1923,$ $P(a_5) = 0.1419.$	$a_1 > a_2 > a_4 > a_3 > a_5$

In contrast, this paper proposes a new measure of interference value based on the idea of grey relational analysis. Grey relational analysis can effectively deal with complex decision-making scenarios with uncertainty and incomplete data, and can more truly reflect the dynamics and complexity of the actual decision-making environment by quantifying the interference effect among decision-makers.

4.3. Sensitivity analysis

In order to observe the influence of the distinguishing coefficient α on the interference intensity among decision-makers, this paper conducts sensitivity analysis on the differentiation coefficient, as shown in Figure 3.

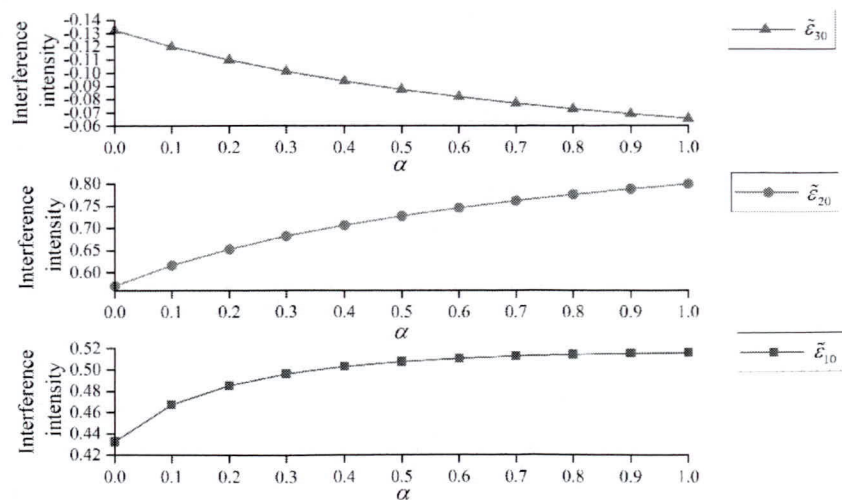


Figure 3 Sensitivity analysis of distinguishing coefficient to interference intensity

As shown in Figure 3, as the differentiation coefficient α changes, the intensity of interference among decision-makers shows different trends. Where, the blue curve represents the tendency of $\tilde{e}_{30}$ to decrease the interference intensity with the increase of α ; The red curve corresponds to $\tilde{e}_{20}$, and the black curve represents $\tilde{e}_{10}$. The interference intensity of both increases with the increase of α . These changes indicate that different decision-makers have different sensitivity to the interference intensity of α .

This phenomenon reflects the nonlinear relationship between the interference intensity and α among decision-makers, revealing the sensitivity of the grey relational analysis model in uncertain group decision-making. In uncertain decision contexts, there are often differences in the evaluations made by decision-makers, which may stem from incomplete information, inconsistent preferences, or environmental uncertainty. The α parameter in the grey relational formula adjusts the weight of the evaluation deviations between decision-makers, particularly the influence of the maximum deviation. When α is large, the model emphasizes the impact of larger deviations, which can lead to an increase in the interference intensity between decision-makers. In this context, higher values of α are better able to reflect extreme opinions or conflicts that significantly affect decision-making, which is particularly important in complex and information-incomplete decision scenarios. On the other hand, when α is small, the model focuses more on the smaller differences between decision-makers, thus reducing the interference intensity between certain decision-makers. This feature, in decision environments characterized by uncertainty and information fuzziness, effectively suppresses the overemphasis on larger deviations between decision-makers, reflects more subtle differences, making it applicable to scenarios where decision-makers' preferences are relatively consistent.

In summary, the variation in α allows the model to flexibly handle different types of uncertainty and interference. By adjusting the weights, it simulates the interactive relationships between decision-makers under various contexts, thus providing dynamic modelling support for the group decision-making process.

5. Case analysis

In this paper, a multi-attribute quantum decision model based on grey relational analysis is proposed to deal with the interaction between decision-makers and improve the applicability and accuracy of the group decision model in complex situations. Firstly, grey relational analysis is used to determine the degree of opinion closeness between the follower decision-maker and the leading decision-maker, and then the weight of decision-maker is calculated. Secondly, a quantum-like Bayesian network model is constructed to incorporate the interference effect among decision-makers into the group decision-making framework, and a method of interference angle calculation based on positive and negative grey relational degree is proposed to improve the accuracy of interference angle measurement between decision-makers. The results show that the model has high accuracy and robustness in complex decision situations with incomplete information and uncertainty.

Although this paper has made some progress in dealing with the problem of mutual interference of decision-makers, there are still some limitations. Firstly, with the increase of the number of decision-makers, the computational complexity of the model is higher, which may affect the efficiency of its practical application in large-scale decision groups. Therefore, future research should focus on the optimization of computational complexity. One potential optimization strategy is to accelerate computation on large data sets by introducing parallel computing frameworks or distributed computing. In addition, heuristic methods or approximate algorithms can be used to simplify the calculation process, so as to improve the practicability of the model. Secondly, the adaptive optimization mechanism in the model is also an important direction of future research. Due to the dynamic nature of decision situations in practical applications, parameters such as decision-maker weights and interference effects in the model may need to be adjusted based on real-time feedback. Therefore, future research can explore how to make the model flexibly adjust in dynamic environment through adaptive algorithms to improve the stability and adaptability of the model in different situations.

In addition, in addition to optimizing computational complexity, future studies can further expand the application field of this model. Although this paper discusses the interference effect in group decision making with the background of emergency rescue, this method is also applicable to other fields and has strong adaptability, especially when dealing with group decision making problems involving uncertainty. It can be further explored to apply this model to medical decision making, financial decision making, supply chain management and other fields, which also involve multi-decision-makers and multi-attribute decision making problems with high uncertainty.

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